

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10: SUB-STRAND 3.2 ROTATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: What do you think makes it possible for a daily pattern of day and night experience on Earth? Use what you learned in all 10 lessons to write your Final Explanation. Answer each section in the space under its prompt. Use the data from our time-lapse video and our lessons (Mombasa, Nairobi, Kisumu and Lake Victoria). Show your working for any calculation. Use the words centre, angle and direction, and write angles in degrees ($^{\circ}$). Remember that positive angles are anticlockwise and negative angles are clockwise. Use East Africa Time (EAT) for times. Write full sentences and give evidence for each claim.

Section 1: Describing the Turning (Centre, Angle, Direction)

In our time-lapse video, 24 hours were shown in 24 seconds, so 1 second = 1 hour. Dawn came over Mombasa at about 6:30 am, midday over Nairobi at about 12:30 pm, sunset over Lake Victoria at about 6:40 pm and darkness over Kisumu at about 7:30 pm, and the pattern repeated every 24 hours. Wanjiku in Kisumu said, 'Kenya feels still.'

(a) Describe Earth's turning using the three things needed to describe any rotation. Give the value or name for each one.

(b) Calculate how many degrees Earth turns in one hour. Show your working.

(c) From the video, the time from dawn over Mombasa (6:30 am) to midday over Nairobi (12:30 pm) is 6 hours. Calculate the angle Earth turns in this time and name the quarter, half or full turn it is.

(a) A rotation is described by a centre, an angle and a direction. Centre: Earth's axis (the line through the North and South Poles). Angle: 15° for every hour (360° in 24 hours). Direction: anticlockwise when viewed from above the North Pole, which is a positive angle.

(b) Earth turns one full turn of 360° in 24 hours. Rate = $360^{\circ} \div 24 \text{ h} = 15^{\circ}$ per hour.

(c) Time = 6 hours. Angle = 15° per hour $\times 6 \text{ h} = 90^{\circ}$. This is a quarter turn, and since it is anticlockwise it is $+90^{\circ}$.

(d) No. Kenya feels still, but it is turning with Earth. Mombasa, Nairobi and Kisumu are carried anticlockwise about Earth's axis at 15° per hour, which is why the video shows a steady turning and the day-night pattern sweeping over Kenya.

(d) Is Kenya really still? Explain using your answers.

Section 2: What Rotation Keeps the Same

In class we rotated triangle ABC with A(2,1), B(4,1) and C(4,3) through 90° anticlockwise about the centre O(0,0).

(a) Use the rule for a 90° anticlockwise turn about O to find the images A', B' and C'.

(b) Find A'' when A(2,1) is rotated through -90° about O. What does the negative sign tell you about the direction?

(c) The sides of ABC are AB = 2, BC = 2 and AC = $\sqrt{8}$. State the lengths of A'B', B'C' and A'C' and say what this tells you about ABC and its image.

(d) Cities such as Nairobi, Mombasa and Kisumu turn with Earth. Use your answer to (c) to explain why they are not stretched or squashed, and what does change.

(a) The rule for 90° anticlockwise about O is $(x,y) \rightarrow (-y,x)$. So A(2,1) $\rightarrow$ A'(-1,2); B(4,1) $\rightarrow$ B'(-1,4); C(4,3) $\rightarrow$ C'(-3,4).

(b) The rule for -90° is $(x,y) \rightarrow (y,-x)$, so A(2,1) $\rightarrow$ A''(1,-2). The negative sign means the turn is clockwise (90° clockwise).

(c) A'B' = 2, B'C' = 2 and A'C' = $\sqrt{8}$. The sides and angles are equal to those of ABC, so the image is directly congruent to the object. Rotation keeps lengths, angles, size and shape the same (it is a direct isometry).

(d) Because rotation preserves lengths, size and shape, Nairobi, Mombasa and Kisumu keep their size, shape and distances from Earth's axis and from each other as they turn together anticlockwise. Only their position and the direction they face change, which is why the pattern repeats regularly.

Section 3: Finding the Centre and Using Symmetry to Explain Regular Turning

(a) A point P(2,0) is mapped to P'(0,2) and Q(4,0) is mapped to Q'(0,4) by a rotation. Construct the perpendicular bisectors of PP' and QQ' and compare them. Can you find the centre from these two? Then a third pair, R(3,2) $\rightarrow$ R'(-2,3), is given. Construct its perpendicular bisector too. Explain what you found about the bisectors, and state the centre and the angle of rotation including its sign.

(a) The centre lies where the perpendicular bisectors of object-image pairs meet, and at least two different bisectors are needed. The bisectors of PP' and QQ' coincide: they are the same line ($y = x$). Two identical lines meet at every point along them, so they do not give one unique intersection and cannot pin down the centre. The bisector of RR' is a different line, and it meets the bisector of PP' at one point, O(0,0). So the centre is O(0,0). Turning P to P' about O is a 90° anticlockwise turn, so the angle is +90°.

(b) Kiondo: order 6, smallest angle = $360^\circ \div 6 = 60^\circ$. Bicycle wheel: order 8, smallest angle = $360^\circ \div 8 = 45^\circ$. Lake Turkana turbine: order 3, smallest angle = $360^\circ \div 3 = 120^\circ$.

(c) The smallest angle of fit = $360^\circ \div \text{order}$, so a full turn is divided into equal steps. In the same way Earth's full turn of 360° in 24 hours is divided into equal steps of $360^\circ \div 24 = 15^\circ$ per hour. Earth's turning is regular and steady, like the equal steps of a wheel, so the cycle repeats evenly every 24 hours.

(b) Find the order of rotational symmetry and the smallest angle of fit for each: Mama Achieng's kiondo pattern (6-fold), Mzee Kamau's bicycle wheel with 8 equally spaced spokes, and a Lake Turkana Wind Power turbine with 3 blades.

(c) Use the idea of dividing a full turn into equal steps to show how it connects to Earth's turning.

Section 4: Why Mombasa Gets Sunrise Before Kisumu

The longitudes are: Mombasa about 39.7° E, Nairobi about 36.8° E and Kisumu about 34.8° E. Earth turns 15° per hour.

(a) Complete this table of longitude differences and the time Earth takes to turn through each difference (give the time to the nearest minute).

Pair of places	Longitude difference	Time to turn
Mombasa and Nairobi	?	?
Nairobi and Kisumu	?	?
Mombasa and Kisumu	?	?

(b) Which of the three places gets sunrise first and which last? Explain using the

(a)

Pair of places	Longitude difference	Time to turn
Mombasa and Nairobi	$39.7^\circ - 36.8^\circ = 2.9^\circ$	$2.9^\circ \div 15 \approx 0.19$ h, about 12 minutes
Nairobi and Kisumu	$36.8^\circ - 34.8^\circ = 2.0^\circ$	$2.0^\circ \div 15 \approx 0.13$ h, about 8 minutes
Mombasa and Kisumu	$39.7^\circ - 34.8^\circ = 4.9^\circ$	$4.9^\circ \div 15 \approx 0.33$ h, about 20 minutes

- (b) Mombasa gets sunrise first, then Nairobi, then Kisumu last. Earth turns anticlockwise (west to east), so places further east turn into the sunlight sooner. Mombasa is furthest east, so it meets the light about 20 minutes before Kisumu.
- (c) Earth turns anticlockwise from west to east, so we are carried towards the east, moving under the Sun's light. Because we do not feel the turning, the Sun appears to move the opposite way, from east to west. The Sun only appears to move; Kenya is turning with Earth.
- (d) We turned the tilted globe (about 23.5°) anticlockwise through 15° for each hour with the torch fixed. The torch lit one half of the globe. The Mombasa dot entered the light first, then Nairobi, then Kisumu.

direction Earth turns. (c) Otieno at Mombasa sees dawn first and Wanjiku at Kisumu sees the Sun set over Lake Victoria. Explain why the Sun appears to move from east to west even though it is Kenya that is turning. (d) In the globe and torch model, what did you do to the globe each hour, and what did you see happen to the dots for Mombasa, Nairobi and Kisumu?	
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Section 5: Final Explanation and Real-Life Uses

Answer the driving question: What do you think makes it possible for a daily pattern of day and night experience on Earth? (a) Write a Final Explanation using a claim, evidence and reasoning. Include centre, angle, direction, the 24-hour cycle, and the Sun lighting one half of Earth. Use at least two pieces of evidence from our lessons, including the Mombasa and Kisumu timing. (b) Key inquiry question 1: How do we use rotation in real-life situations? Describe how a clock and a wind turbine at Lake Turkana are described with rotation. For the clock give the angle the minute hand turns in one minute and the angle the hour hand turns in one hour, with signs. State the angle a turbine blade must turn to land on the next blade's position. (c) Key inquiry question 2: Where do we apply rotational	(a) Claim: Day and night repeat every 24 hours because Earth rotates once (360°) about its axis in 24 hours, turning anticlockwise at 15° per hour as seen from above the North Pole, while the Sun always lights one half of Earth. Evidence: In the time-lapse (24 hours in 24 seconds), dawn came to Mombasa at about 6:30 am, midday to Nairobi at about 12:30 pm, sunset over Lake Victoria at about 6:40 pm and darkness over Kisumu at about 7:30 pm, and the pattern repeated every 24 hours. With the globe and torch, turning the globe 15° each hour brought Mombasa into the light first, then Nairobi, then Kisumu. Mombasa and Kisumu differ by 4.9° of longitude, and $4.9 \div 15$ is about 20 minutes. Reasoning: The turning is described by a centre (Earth's axis), an angle (15° per hour) and a direction (anticlockwise). As Earth turns, each place moves from the lit half into the dark half and back again, so day and night follow each other regularly. Mombasa is further east, so it turns into sunlight about 20 minutes sooner than Kisumu. The Sun only appears to move from east to west; Kenya is not still, because it turns with Earth. (b) The minute hand turns $360^\circ \div 60 = 6^\circ$ per minute, and the hour hand turns $360^\circ \div 12 = 30^\circ$ per hour. Clock hands turn clockwise, so the angles are -6° per minute and -30° per hour. The centre is the pin in the middle of the clock. A Lake Turkana turbine has 3 blades, so it has order 3, and a blade must turn $360^\circ \div 3 = 120^\circ$ to land on the next blade's position. Its turning is described by a centre (the hub), an angle and a direction, and rate $\times$ time gives the angle turned, just as for Earth (15° per hour $\times$ time). (c) Examples: Mama Achieng's kiondo pattern has order 6 (60°), Mzee Kamau's 8-spoke bicycle wheel has order 8 (45°), and a matatu wheel with 5 nuts has order 5 (72°). (d) Any two of: label Earth's axis as the centre of rotation; show the angle of 15° per hour and an anticlockwise arrow as seen from above the North Pole; show that Mombasa turns into the light about 20 minutes before Kisumu; show that the Sun only appears to move east to west; show that Kenya turns with Earth.
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symmetry in day-to-day life? Give three examples and state the order of each. (d) Look back at your first model from Lesson 1. State two things you would add or change to your model now.	
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Describing rotation with centre, angle and direction	Correctly names Earth's axis as the centre, 15° per hour (with the $360^\circ \div 24$ working) as the angle, and anticlockwise (positive) as the direction, and uses them consistently, including the 90° in 6 hours calculation.	Names the centre, angle and direction with only one small error or missing working, for example omits the sign of the angle.	Names fewer than three of the elements or gives wrong values, for example confuses clockwise and anticlockwise or gives the wrong rate.
Properties of rotation and congruence	Correctly finds A', B', C' and A'' using the rules, states the sides of the image, and explains that rotation preserves length, angle, size and shape (direct congruence), with only position and facing changing.	Finds most images correctly and states that the shape is preserved but gives only a partial explanation linked to the cities.	Makes errors with the coordinate rules or the sign of the angle, or does not link preserved shape and size to the congruence idea.
Calculations with centre finding, symmetry and longitude timing	Explains that the first two bisectors coincide and so give no unique intersection, uses a third bisector to find the centre O(0,0) and $+90^\circ$, gives correct orders and angles using $360^\circ \div \text{order}$, and a fully correct table (2.9° , 2.0° , 4.9° giving about 12, 8 and 20 minutes).	Most values are correct with one or two arithmetic or rounding slips, or reasoning for the third bisector is incomplete.	Several values are incorrect or missing, or the method (dividing by 15 or $360 \div \text{order}$) is not shown.
Claim-evidence-reasoning explanation of the phenomenon	States a clear claim that Earth rotates once about its axis in 24 hours with the Sun lighting one half, supports it with at least two specific pieces of evidence from the lessons (video times, globe and torch, longitude difference), and reasons that Mombasa gets sunrise about 20 minutes before Kisumu and that the Sun only appears to move east to west.	States a correct claim with one piece of evidence and reasoning that is mostly complete.	Gives a claim without evidence or reasoning, or includes a misconception such as the Sun moving around a still Earth.
Real-life applications and model revision	Correctly describes the clock (-6° per minute, -30° per hour) and the 3-blade turbine (order 3, 120°), gives three	Gives correct applications with one error, or only one clear improvement to the	Applications are missing or wrong, or no clear change to the model is described.

	correct symmetry examples with orders, and identifies two specific improvements to the Lesson 1 model.	model.	
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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